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# An embedded system for AC series arc detection by inter-period correlations of current

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ABSTRACT

This paper presents a method for detecting series arcs for a 230 V AC–50 Hz residential installation, its implementation in an embedded circuit and its performance under real conditions. First, we evaluated the inter-period autocorrelation coefficient *r* of the line current. The algebraic estimation of the derivative of the signal *r* was then used to obtain an indicator of arc detection. *r* was calculated on the current at a sampling rate frequency of 5 kHz and the 50 Hz component was attenuated by an high pass (HP) filter. Tests carried out with domestic appliances with different power loads (resistive load, lamp, vacuum cleaner, dimmer etc) show a value for *r* which is very stable and close to 1 under normal conditions. The coefficient *r* weakens in the presence of an arc (initiated by a carbon path) and is no longer constant. The complete algorithm was implemented in real time on a 32-bit ARM Cortex-M3 microcontroller with a 12-bit ADC. The measuring current, which ranges from 0 to more than 20 A, was provided by a passive toroidal inductive probe. Four typical household loads were tested. The results obtained show that detection is operative and very fast (10 ms) with currents which exceed 400 mA.

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## 1. Introduction

Developing reliable arc fault detectors is a major issue in protecting AC domestic installations. Indeed, in the United States and Europe, several studies have shown that a significant proportion of domestic fires are caused by arcing faults [1,2]. Unlike the United States, where these circuits are imposed in homes, this is not the case in the Europe (EEC). However, new European Regulations (IEC 62606) [3] will make them compulsory in the coming years. Currently, some devices are available which are suitable for detection in 230 V/50 Hz AC power installations.

The main methods for detecting arc faults, as presented in current literature, are based on an analysis of the line current. Several detection methods are based on current spectral composition. The analysis is performed in a variety of bands ranging from low frequencies up to 20 kHz [4–7] and at even higher frequencies [8,9]. Other methods focus specifically on analyzing the odd harmonics [10], more specifically third and fifth harmonics [4,5,11]. In general the frequency range is selected to avoid interference caused by the load. Other solutions operate with a time domain feature approach—rapid changes in arcing currents ($di/dt$) [12], autocorrelation [13], crest factor or inter-harmonic distortion [14,15]. Some methods are based on the implementation of time-frequency analysis or wavelet packet transform [16], which require long and empirical settings which negatively affect their uses on a large scale.

Siemens, for example, proposes a detection device for the European market for use with low and average AC voltage (AFD Siemens unit [17]) which is suitable for currents between 3 and 16 amps.

Very few studies show detection results in the presence of nonlinear or switching (dimmer) loads where the power signature is closed to an arc signature [18]. Finally, it is difficult to find studies giving the lower current limit above which the detection method is reliable.

In a previous paper [19], we presented a detection method based on the algebraic derivation of the line current. This method, which can be used in AC or DC regimes, is very robust with respect to measurement noise, but remains limited to installations which do not use switching electrical devices. It is therefore not possible to apply the algebraic derivation directly on the current in domestic electrical installations where many domestic loads include dimmers.

To overcome this limitation, we looked for an arc indicator which is far less sensitive to the nature of the load than the current itself. Numerous studies [20–22] have demonstrated the non-stationary nature of the statistical characteristics of the current in the presence of a series arc. The random behavior of the arc introduces temporary differences between successive periods which can be demonstrated by measuring the correlation coefficient from one period to another. The detection strategy presented here is based on the monitoring of real-time fluctuations of this coefficient by algebraic derivation.

In the first part of this paper, we give the assumptions made in our study and present the theoretical background based on the calculation of the inter-period autocorrelation coefficient $r$ from the current line and its derivative by an algebraic estimator. Off-line, the calculation of $r$ is performed on real currents in the presence or absence of a series arc. The results of the calculations are summarized in the second part. The third and final parts are devoted to the implementation of the detection method on an embedded operating system produced around an ARM Cortex-M3 microcontroller.

## 2. Method

### 2.1. Theoretical background

The principle that we propose derives from the double hypothesis that first, the statistical characteristics of the line current in the absence of arc are constant throughout a period $T_0$ (the load can be linear or not), and that second, the arc phenomenon results from a non-stationary random process [20–23].

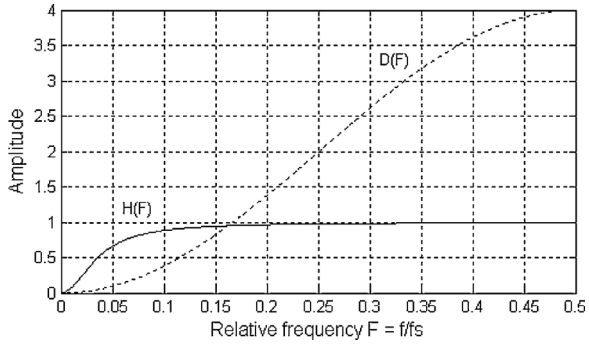



**Fig. 1.** Time frequency responses of the notch filter and second order differentiator.

In order to produce an arc discharge between two electrodes, after the zero-crossing of the voltage and current, the increase in voltage will cause ionization of the isolator, which must be broken when the voltage reaches the restrike value. This value may vary greatly from one cycle to the other, and depends on numerous factors: gap width, the surface condition of the electrodes and the temperature and presence of ionized air [22]. Therefore the restrike of arc voltage behaves randomly.

In practice, we observe an inter-period correlation coefficient $r$:

- Stable and close to 1 in normal operation.
- Lower and variable with the arc presence.

However, in order to highlight the fall of $r$, it is necessary to attenuate the deterministic components of the current, especially the fundamental frequency. The DC component (offset of the measuring probe and bias circuit) must be totally eliminated.

To meet these constraints, the current $I$ is filtered by a notch filter centered on zero frequency. Its transfer function is given by:

$$H(z) = (1-e)\frac{1-2z^{-1}+z^{-2}}{1-2(1-e)z^{-1}+(1-e)^2z^{-2}} \quad (1)$$

This filter acts as a high pass (HP) filter with a low cutoff frequency and a very narrow rejected band. $e$ is a positive real value well below 1 and the −3 dB cutoff frequency ($f_{cutoff}$) is obtained according to the following equation:

$$f_{cutoff} \approx \frac{e}{4} f_s \quad (2)$$

with $f$ the signal frequency and $f_s$ the sampling frequency.

Nevertheless, the fundamental component (50 Hz) should not be completely eliminated in order to allow the correct calculation of $r$ for little deformed current. This is the case for linear loads without the presence of an arc in the line.

The numerator of $H(z)$ is identical to a second order differentiator function. The notch filter is adapted to highlight discontinuities generated by the arc current, while being less sensitive to high frequency noise.

The frequency responses of the notch filter (where $e = 0.2$) and the second order differentiator are compared in Fig. 1.

The autocorrelation coefficient $r_k(n)$ can be estimated at each sampling time using the Pearson formula [24]:

$$r_k(n) = \frac{\sum_{j=1}^{M}\left[\left(x_{k-j}-\bar{x}_k\right)\left(x_{k-n-j}-\bar{x}_{k-n}\right)\right]}{\sqrt{\left[\sum_{j=1}^{M}\left(x_{k-j}-\bar{x}_k\right)^2\right]\left[\sum_{k=1}^{M}\left(x_{k-n-j}-\bar{x}_{k-n}\right)^2\right]}} \quad (3)$$

where $x_k$ is sampled and filtered current signal.

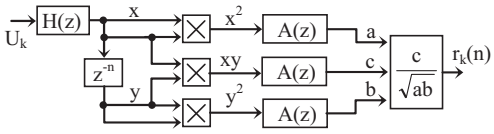



**Fig. 2.** Real time algorithm for $r$ calculus.

The inter-period autocorrelation coefficient is obtained when $n$ is equal to $n_0$.

Where $n_0 = f_s/f_0$ represents the number of samples in a fundamental period of the current $T_0 = 1/f_0$.

The statistical parameter $r$ is theoretically insensitive to the amplitude $x$ of the considered signal. Its definition domain is the interval [−1 1].

The formulation (3) is not adapted to the calculus on a sliding window. We prefer the equivalent expression (4) that separates all mean data.

$$r_k(n) = \frac{\overline{x_k x_{k-n}} - \bar{x}_k \cdot \bar{x}_{k-n}}{\sqrt{\left(\overline{x_k^2} - \bar{x}_k^2\right)\left(\overline{x_{k-n}^2} - \bar{x}_{k-n}^2\right)}} \tag{4}$$

As the signal $x$ has a zero DC component ($\bar{x}_k$ and $\bar{x}_{k-n} = 0$), (4) is reduced to:

$$r_k(n) = \frac{\overline{x_k x_{k-n}}}{\sqrt{\overline{x_k^2}.\overline{x_{k-n}^2}}} \tag{5}$$

### 2.2. Algorithm for r calculus

The diagram which provides the calculation of $r$ in real time at the rate $f_s$ is shown in Fig. 2.

$U_k$ is the sampled voltage delivered by the current probe.

$A(z)$ is the transfer function of the recursive average filter upon $M$ terms:

$$A(z) = \frac{1}{M}\frac{1 - z^{-M}}{1 - z^{-1}} \tag{6}$$

The integer $M$ is set to a value greater than or equal to $n_0$.

### 2.3. Detection based on the algebraic derivative of r

The proposed detection is based on monitoring the level of $r(n)$ and its variations. The coefficient $r$ being always inferior or equal to 1, the simplest solution is to set a threshold below which the arc indicator is activated. In practice, for certain loads and in absence of an arc, it is not impossible to see a coefficient $r(n_0)$ less than 0.9. Therefore the choice of a single threshold is problematic.

The passage of $r$ from a quasi-static state to a variable regime constitutes a more reliable indicator. Whatever the adopted indicator, the variations of $r$ due to on/off states, load changes and so on, should be inhibited (holdoff) to avoid false detections.

To detect signal regime change, we propose to apply an algebraic differentiator. An algebraic differentiator is developed for estimating the derivative of a signal $f(t)$ from the limited development of Taylor–McLaurin transposed in Laplace space.

For example, for a limited second order development:

$$f(t) = f(0) + \frac{f'(0)}{1!}t + \frac{f''(0)}{2!}t^2$$

This gives, after Laplace transform:

$$p^3F(p) = p^2f(0) + pf'(0) + f''(0)$$

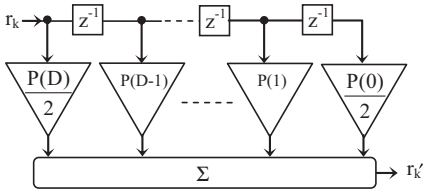



**Fig. 3.** Structure of the estimator filter.

By isolating $f'(0)$ and after return in the temporal space, an estimator of the first derivative is obtained according to the integrated expression:

$$f'(0) \approx -\frac{24}{t^4}\int_0^t \left[\frac{3}{2}(t-\tau)^2 - 5\tau(t-\tau) + \tau^2\right] f(\tau)d\tau \tag{7}$$

To obtain a real-time estimation, integration must be performed over a sliding window of duration $T$ which is to replace $t$ by $T$ in expression (7).

For the sampled signal $r_k$, the integration window has a width $T = D \times T_s$ where $D$ is a selected integer.

The integration variable can be written as: $\tau = d \times T_s$ with $0 \le d \le D$

The digital version of (7) can be synthesized by an FIR filter with $(D+1)$ coefficients from the polynomial $P(d)$:

$$P(d) = G\left[\frac{3}{2}(D-d)^2 - 5d(D-d) + d^2\right] \tag{8}$$

With $0 \le d \le D$ and $G = -\left(24/D^4\right) f_s$

Fig. 3 shows the structure of the algebraic derivative filter. The following features comprise the structure of the filter:

- The end filter coefficients are halved to produce a numerical integration according to the trapezoidal method;
- The integration step $T_s$ is taken into account in the factor $G$;
- $r'_k$ is expressed in $s^{-1}$;
- The numerical integration is accurate – even more accurate when $D$ is higher. It may be necessary to adjust the central coefficient $P(D/2)$ in order for the sum of the filter coefficients to be equal to zero;
- Further explanations of this method are given in [19].

## 3. Off line correlation coefficient calculus

### 3.1. Experimental conditions

The electrical circuit (Fig. 4), designed to evaluate the performance of the detection method, consists of a 4 kW generator that supplies various domestic charges in an alternating voltage of 230 V, $f_0 = 50$ Hz. According to the IEC62606 standard, the ignition of series arc is performed using a carbon path (CP) inserted in the circuit.

The complete experimental setup is given in Fig. 5 [25]. Current measurements are made using a wideband Hall effect sensor (Lecroy AP015) (1). Measurements are sampled at $f_s = 5$ kHz, digitized and stored on 12 bits using an oscilloscope Lecroy HDO 6104 (2).

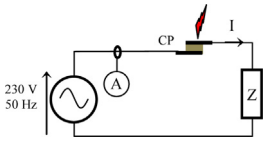



**Fig. 4.** Experimental electrical circuit.

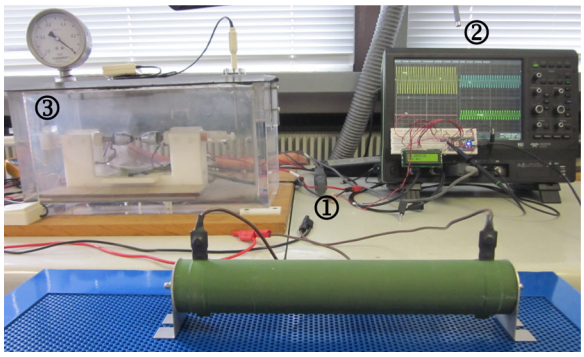



**Fig. 5.** Experimental test bench.

**Table 1**
Loads under test.

| | Load type | Symb. | $I_{RMS}$ (A) | Power (W) |
|---|---|---|---|---|
| **Test 1** | Resistance | R | 4.9 | 1125 |
| **Test 2** | Motor | M | 5.5 | 1250 |
| **Test 3** | Dimmer | D | 4 | 900 |
| **Test 4** | Combined loads | C | 6.8 | 1500 |

The arcs are created in a sealed chamber (3) in which we can measure pressure and internal temperature.

The CP is formed by juxtaposing two insulating electric cables which have been damaged with a cutter. The ignition is provided with high voltage [3].

The nature and characteristics of domestic loads used are summarized in Table 1.

The universal motor is used in a Philips vacuum cleaner. The combined load is formed by the parallel association of the dimmer with a 600 W resistor. Recordings are made over a period of about 2 s (10,000 points). They comprise a first zone from 0 to 1 s without arc (normal operation) and a second zone from 1 to 2 s in which an arc is present.

### 3.2. Typical currents

The currents $I(t)$ and filtered signals $x(t)$ are presented in Figs. 6–9. The filter parameter $e$ was set to 0.2.

The 50 Hz component is attenuated by about 23 dB and the filter bandwidth begins at 250 Hz.

In order to visualize the signal shape, only 8 periods (4 without and 4 with an electric arc) are reproduced.

As expected, the notch filter accentuates the discontinuities and clearly shows the randomness of the disturbances introduced by the arc.

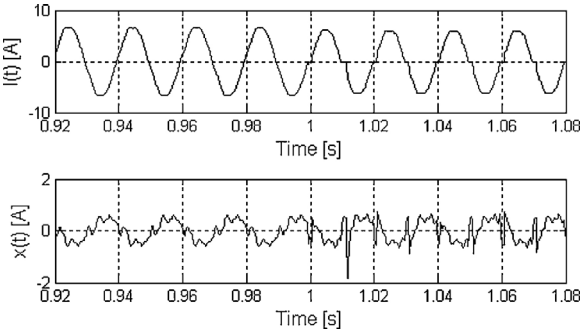



**Fig. 6.** Current in the resistance (before and after filtering).

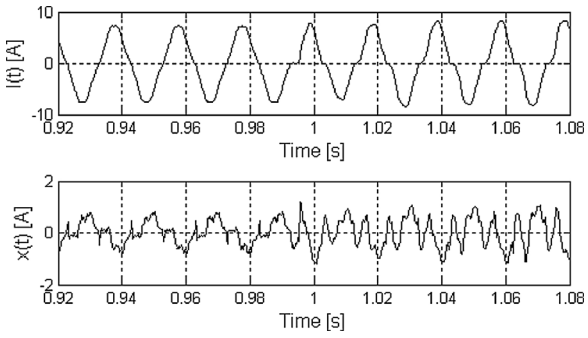



**Fig. 7.** Current in the motor (before and after filtering).

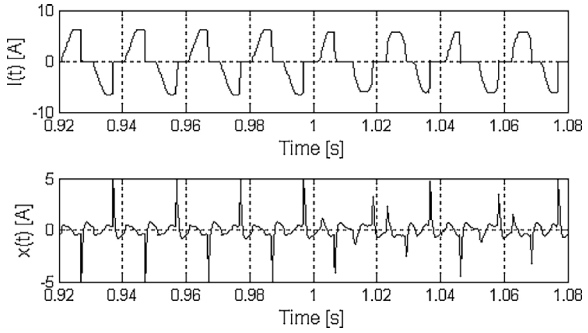



**Fig. 8.** Current in the dimmer (before and after filtering).

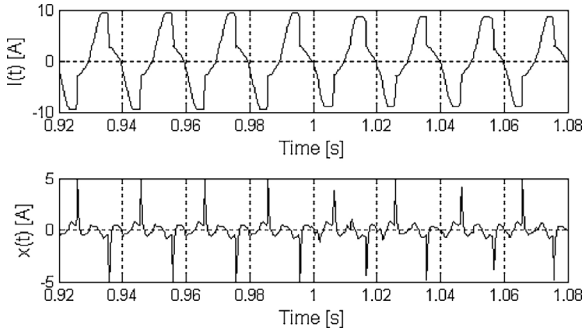



**Fig. 9.** Current in the combined load (before and after filtering).

### 3.3. Off line simulation

We simulated the algorithm by calculating $r$ using a Simulink model according to Fig. 2 with the following parameters $n = M = 100$ and $e = 0.2$.

The evolution of $r$, in Figs. 10 and 11, is obtained for each load, with and without arc.

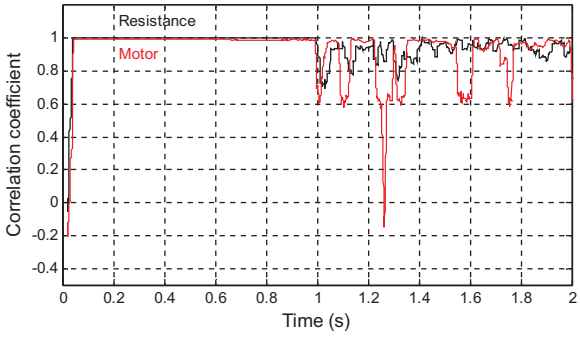



**Fig. 10.** Evolution of $r$ coefficient for the resistance and the motor.

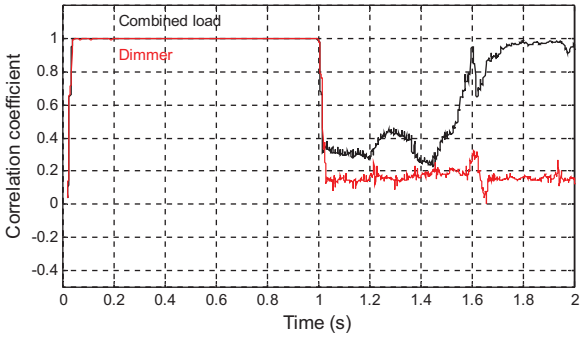



**Fig. 11.** Evolution of $r$ coefficient for the dimmer and the combined load.

The value $n = 100$ (delay on a period) guarantees, in the absence of arcing, a stable correlation coefficient practically equals to 1, whatever load is used.

The presence of a series arc can be detected unambiguously as soon as it appears, whatever the load. The temporal evolution of $r$ is almost horizontal in normal operation and has strong variations in the presence of arc.

Rapid changes of $r$ have durations of a few milliseconds which generate peaks from the derivate of order 10 to 30 $s^{-1}$.

### 3.4. Comparison with other arc indicators

Different indicators of varying complexity have been proposed by other teams. For example, indicators based on the crest factor [14] and the variance [18] of the current offer a good compromise between reliability and the possibility of real-time implementation. We compared the evolution of the coefficient $r$ to these indicators by applying to them the same current measured in the dimmer. The comparative results are shown in Fig. 12.

The stability of an indicator can be defined by its standard deviation $\sigma$ operating without arc. The lower it is, the more the risk of false detection is minimized. For the coefficient $r$, $\sigma$ is stabilized to $3.86 \times 10^{-4}$. This value is much lower than that obtained for the other two indicators ($75 \times 10^{-4}$ for the creast factor and $83 \times 10^{-4}$ for the variance).

The sensitivity of an arc indicator can be estimated by the amplitude of its variations in the presence of an arc compared to the baseline without the arc. We observe significant level changes (2.03 dB and −1.36 dB) for the crest factor, but the sign is inversed from one half period to the next. The average level fall is stabilized at −1.46 dB for the variance and at −15.68 dB for the coefficient $r$. Although further tests are needed to confirm this finding, it seems that the inter-period autocorrelation coefficient performs better in terms of sensitivity and stability.

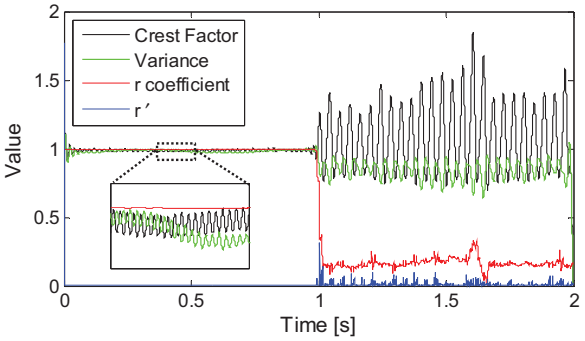



**Fig. 12.** Comparison between three arc indicators.

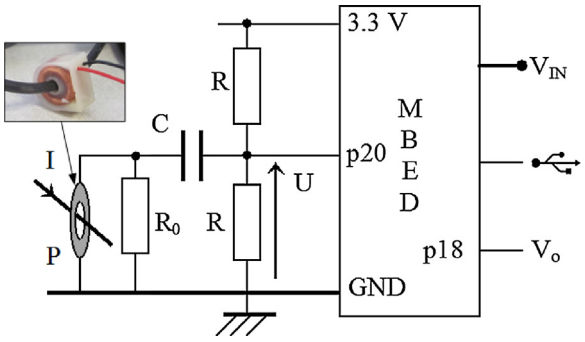



**Fig. 13.** The detector electrical diagram.

## 4. Embedded detector

### 4.1. Hardware

We have developed our system around the MBED microcontroller LCP1768 which is particularly well suited for rapid prototyping of embedded electronic systems.

The LCP1768 board includes an ARM Cortex-M3 processor bits/100 32 MHz, 64 kB SRAM, 512 kB of flash memory, 12-bit ADC and 10-bit DAC. It has different communication ports and a USB type 2 MB flash memory, very useful for recording special events during operation.

After compilation, the programs are transferred to MBED via the USB port.

The LCP1768 board can be powered by a voltage (V) between 4.5 V and 9 V. It has an internal voltage regulator of 3.3 V.

Representing the electronic detector diagrammatically is simple, being limited to the microcontroller, an inductive sensor current and some passive components (Fig. 13).

In order to make gains in mobility, the supply voltage $V_{IN}$ can be ensured by a 4.8 V NiMH battery.

The probe P is a toroidal inductive coil (Fig. 13). Loaded with a resistor $R_0 = 100\,\Omega$, its sensitivity is approximately 70 mV/A at 50 Hz.

The signal delivered by the probe P is applied by a capacitive connection ($C = 10\,\mu F$) to the input (p20) of the ADC. The latter is polarized at 1.65 V by two resistors $R$ (10 kΩ).

Given the supply voltages, the dynamic range of current measurement $I$ extends from 0 to 23 A.

The different test results can be stored in an internal file, transmitted by a digital port or as an analog voltage ($V_0$).

A photograph of the electronic device (MBED and LCD display coefficient $r$) is shown in Fig. 14.

The main program, developed in C++, synthesizes the algorithm given in Fig. 2. We provided some additional routines performing the LCD display, internal storage and output ($V_0$) of the coefficient $r$ or its derivative $r'$.

To avoid multiple memory accesses that may affect the real-time operation of the program, different delays ($z^{-n}$ and $z^{-M}$) are carried out using circular buffers.

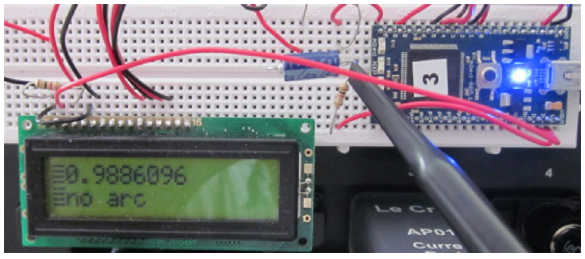



**Fig. 14.** Embedded detector.

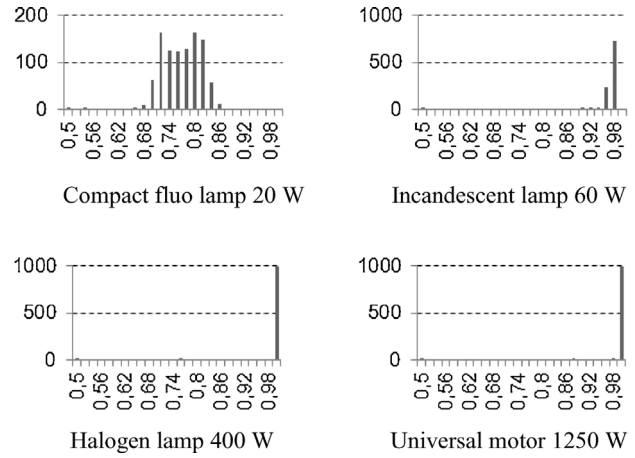



**Fig. 15.** Dispersion of $r$ for some domestic loads.

The sampling rate $f_s$ of 5 kHz is handled by an interrupt function. And in all the following tests $M = n = 100$.

### 4.2. Stability of the coefficient r in the absence of an electric arc

In order to verify the stability of the correlation coefficient, we identified 1000 values for $r$ delivered by the detector for different domestic loads during normal operation, over a period of about 15 min.

The dispersion of $r$ for several loads is represented in Fig. 15.$r$ has a value greater than 0.94 when the current flowing through the load exceeds the RMS value of 250 mA (load of 60 W). For currents greater than 1 A, $r$ shows a value of $0.99 \pm 0.01$.

In contrast, for currents less than 100 mA (load power less than 20 W), the stability of $r$ is no longer guaranteed in normal operation. This lower limit is imposed by the noise generated by the toroidal sensor and the quantization precision of the ADC.

### 4.3. Limit of detection versus the current I

We periodically applied the signals used in off-line tests to verify the correct operation of the arc detector. This was done by using an arbitrary waveform generator (Agilent 33500B) in which files were imported signals. By so doing, the detection performances are easier to evaluate than with an isolated electric arc, which is difficult to reproduce.

The output voltage $V_0$ of the DAC is used to display the values of $r$ or $r'$ on an oscilloscope.

For example, Fig. 16 shows the voltage $V_0$ corresponding to the autocorrelation $r$ calculated in real time on the measured current in the motor signal. The relation $V_0(r)$ follows the expression:

$$V_0(r) = 3.3 \cdot r \quad [\mathrm{V}]$$

The evolution of $V_0(r)$ is identical to the calculus of $r$ represented in Fig. 10.

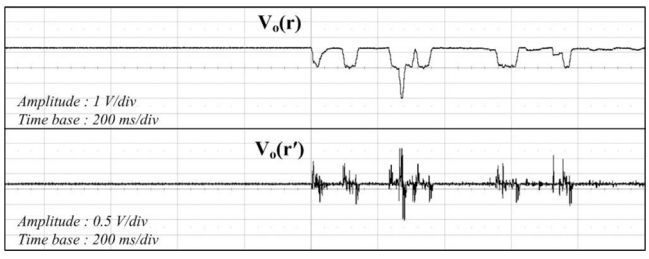



**Fig. 16.** Oscillogram of $V_0(r')$ for the motor.

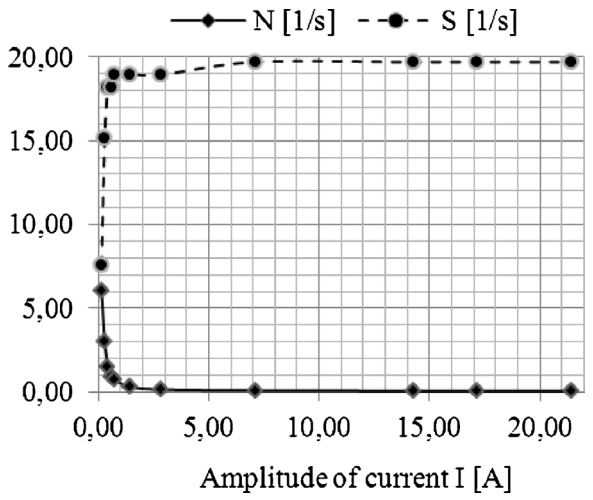



**Fig. 17.** Level of noise ($N$) and derivative peaks ($S$) versus the current $I$ amplitude.

For the estimation of the derivative $r'$, we sized the number $D$ of FIR filter coefficients so that the duration of integration is of the same order as that for the most rapid variation of $r$.

For $D$ equals 10, the period integration is 2 ms which is appropriate; by choosing this, the detector's response time is slightly increased.

The algebraic polynomial differentiator implemented in the microcontroller is then written as:

$$P(d) = -6 \cdot \left[3D^2 - 16dD + 15d^2\right] \tag{9}$$

with $0 \le d \le 10$

The algebraic method has been reported previously by the authors in [21].

To visualize the evolution of $r'$, we use the analog scale setting:

$$V_0(r') = 3.3 \left[0.5 + \frac{r'}{50}\right] \quad [\mathrm{V}] \tag{10}$$

This allows us to view peaks for the derivative up to $25\,\mathrm{s}^{-1}$, before saturation of the DAC.

The evolution of $V_0(r')$, relative to the measured current in the motor, is reproduced in Fig. 16.

In order to check its influence on the detector performance, the amplitude of the signal U can easily be modified with the arbitrary shape generator.

We varied the amplitude $U$ from 0 to 1500 mV, which corresponds to a measured current of from 0 to 21.5 A, by using the measured motor current.

We measured the digital signal $r'$ each time:

- The mean noise level $N$ (baseline) in the absence of an arc;
- The average amplitude $S$ of the peaks in the presence of an arc.

Fig. 17 shows the changes of $N$ and $S$ depending on the current amplitude $I$.

As soon as the current exceeds 0.4 A, the presence of an arc is indicated with a difference greater than 20 dB compared with the baseline. This contrast reaches 50 dB for currents above 10 A.

By comparing the digital values $r'$ to a threshold of $2\,\mathrm{s}^{-1}$, it is possible to reliably detect the presence of an arc for currents whose amplitude exceeds 0.4 A. At lower currents, the risk of non detection or false alarms is expected.

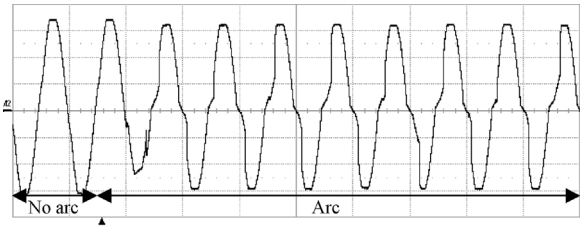



**Fig. 18.** Current $I$ in the circuit (4.9 A RMS).

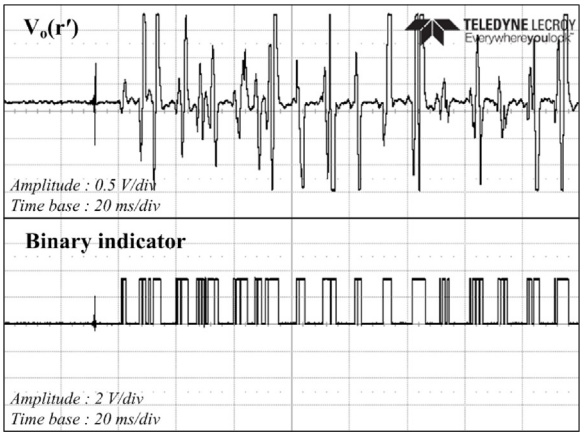



**Fig. 19.** Derivative coefficient $r$ and binary indicator of the arc presence.

### 4.4. Time response of the arc detector

Before concluding, we present a final test performed in a real situation on a circuit comprising a resistive load of 47 Ω in series with a CP.

This test was designed to evaluate the time response of the detector.

Initially the CP was short-circuited by a closed switch. It was then opened to arc ignition at the required time.

Fig. 18 represents the current in the circuit. Arc ignition time is indicated on the abscissa by a triangular index.

Variations in the coefficient $r$ are related with the appearance of a current distortion. The only operations that cause delays are the averaging filter and the algebraic estimator.

We can theoretically estimate the response time of the detection algorithm by the relationship:

$$\tau_{\text{th}} = \left(\frac{M}{2+D}\right) T_s \tag{11}$$

The signal $V_0(r')$ and the binary state obtained for a detection threshold of $2\,\text{s}^{-1}$ are shown in Fig. 19.

The first disturbance detected on $r'$ is generated by the opening of the switch.

These oscillograms indicate a delay of about 10 ms between the ignition time of the arc and the first indication of presence of the arc.

This experiment confirms the rapidity of the arc detector where the response time satisfied the IEC62606 standard which imposes a maximum time response of 500 ms for a current of 5 A RMS and 250 ms for a current of 10 A RMS.

## 5. Conclusion

Traditional arc detection solutions are based, for the most part, on spectral analysis of the line current. The method we propose is different and is original in fundamentally exploiting the random behavior of the arc phenomenon. The statistic parameter that we use is the normalized autocorrelation inter-period coefficient $r$ of the current. The current is first filtered by a notch filter centered on zero frequency which, by attenuating the fundamental component $f_0$, highlights the differences between two successive periods. The coefficient $r$ has the advantage of being very sensitive to the current amplitude when it exceeds the noise level. Our tests show a normal value of $r$ which is very stable and above 0.94 for currents above 250 mA. In the presence of an arc and irrespective of the load (linear or switching) the coefficient $r$ weakens and is no longer constant.

The detection algorithm includes two main modules. The first calculates the coefficient $r$ and the second is an FIR filter which estimates the derivative of $r$ in real time. The estimation is performed by an algebraic approach that is very resistant to noise. The time response of the algorithm, approximately half a current period (10 ms in Europe), largely exceeds the requirements imposed by the various standards.

The detection method is currently implemented on an embedded system based around an ARM microcontroller. Current measurement is made by a single toroidal passive probe. Capable of being operated with currents of low amplitude and at moderate cost, this type of detector can be applied to a large number of electrical safety applications.

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## References

[1] FEMA, Topical Fire report series: residential building electrical Fires United States Fire Administration, 8, 2008.
[2] Groupe de Réflexion sur la sécurité électrique dans le logement (GRESEL). La sécurité électrique. (2014). Available: http://www.gresel.org/incidentelectrique-2-4.html
[3] IEC62606, General requirements for arc fault detection devices, first ed., International Electrotechnical Commission, March 2013, pp. 1–150.
[4] N. Hadziefendic, M. Kostic, Z. Radakovic, Detection of series arcing in low-voltage electrical installations, Eur. Trans. Electr. Power 19 (2009) 423–432.
[5] P. Muller, S. Tenbohlen, R. Maier, M. Anheuser, Characteristics of series and parallel low current arc faults in the time and frequency domain, in: 56th IEEE Holm Conference on Electrical Contacts (HOLM), 2010, pp. 1–7.
[6] P. Dae-won, K. Il-kwon, C. Su-yeon, K. Gyung-suk, Detection algorithm of series arc for electrical fire prediction, in: International Conference on Condition Monitoring and Diagnosis, Beijing, 2008, pp. 716–719.
[7] B.M. Aucoin, B.D. Russell, Distribution high impedance fault detection utilizing high frequency current components, IEEE Trans. Power Apparatus Syst. PAS-101 (1982) 1596–1606.
[8] X. Zhou, J.J. Shea, K.L. Parker, and T. J. Miller, Arc fault circuit interrupter and series arc fault detection method using plural high frequency bands, U.S. Patent 7 633 727, (2009).
[9] A. A. Haun, B.G. Grattan, K.B. Wong, R.F. Dvorak, and G. W. Scott, Arc fault circuit interrupter, U.S. Patent 6 414 829, (July 2002).
[10] B.D. Russell, R.P. Chinchali, C.J. Kim, Behaviour of low frequency spectra during arcing fault and switching events, IEEE Trans. Power Delivery 3 (1988) 1485.
[11] D.C. Yu, S.H. Khan, An adaptive high and low impedance fault detection method, IEEE Trans. Power Delivery 9 (1994) 1812–1821.
[12] K. Phipps, T. Cooke, D. Dorr, P. Keebler, Frequency phenomenon and algorithms for arc detection, in: IEEE International Symposium on Electromagnetic Compatibility (EMC), 2010, pp. 183–188.
[13] J. Lezama, P. Schweitzer, S. Weber, E. Tisserand, P. Joyeux, Arcing detection at home system using correlation analysis, in: The 27th International Conference on Electrical Contacts, 2014, pp. 1–6.
[14] C. Kim, B. Russell, Analysis of distribution disturbances and arcing faults using the crest factor, Electr. Power Syst. Res. 35 (1995) 141–148.
[15] B.D. Russell, R.P. Chinchali, C.J. Kim, Behaviour of low frequency spectra during arcing fault and switching events, IEEE Trans. Power Delivery 3 (1988) 1485–1492.
[16] K. Koziy, G. Bei, J. Aslakson, A low-cost power-quality meter with series arc-fault detection capability for smart grid, IEEE Trans. Power Delivery 28 (2013) 1584–1591.
[17] Restrepo, P. Staley, A. Nayak, V. Mikani, H. Kinsel, S.R. Titus, et al., Systems and methods for arc fault detection, U.S. Patent 7,864,492, (Jan 2011).

[18] J. Lezama, P. Schweitzer, S. Weber, E. Tisserand, P. Joyeux, Arc fault detection based on temporal analysis, in: 60th IEEE Holm Conference on Electrical Contacts (Holm), 12–15 Oct 2014, 2014, pp. 1–5.
[19] E. Tisserand, J. Lezama, P. Schweitzer, Y. Berviller, Series arcing detection by algebraic derivative of the current, Electr. Power Syst. Res. 119 (2015) 91–99.
[20] S.-C. Wang, C.-J. Wu, Y.-J. Wang, Detection of arc fault on low voltage power circuits in time and frequency domain approach, Int. J. Circuits Syst. Signal Process. 6 (5) (2012) 324–331.
[21] N. Zamanan, J.K. Sykulski, Modelling arcing high impedances faults in relation to the physical processes in the electric arc, WSEAS Trans. Power Syst. 1 (2006) 1507–1512.
[22] G. Parise, L. Martirano, M. Laurini, Simplified arc-fault model: the reduction factor of the arc current, IEEE Trans. Ind. Appl. 49 (July–Aug (4)) (2013) 1703–1710.
[23] C.J. Kim, An arc/spark detection system as an early warning indicator of electric fire hazard—final report for the project, in: Development of a Home Networking System for Automation and Security Enhancement, Howard University, Washington, USA, Office of Research Administration, 2002 (June).
[24] J. Hine, G.B. Wetherill, Coefficient of correlation, in: A Programmed Text in Statistics Book 4: Tests on Variance and Regression, Springer, Netherlands, 1975, pp. 32–41.
[25] J. Andrea, P. Schweitzer, E. Tisserand, P. Roth, S. Weber, Calibrated AC and DC Arcing Fault Generator, in: 56th IEEE Holm Conference on Electrical Contacts (HOLM), 2010, pp. 1–6.